CASCADIA ANALYTICS

Taiwan Semiconductor Manufacturing Co

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Priced Beyond Perfection as Extreme Capital Intensity Compresses Cash Conversion

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Taiwan Semiconductor Manufacturing Co maintains an unassailable operational chokehold over leading-edge silicon manufacturing, delivering record top-line expansion fueled by high-performance computing acceleration. However, public market equity valuations at the reference spot price of $432.75 reflect an unsustainable extrapolation of peak operating margins, discounting neither the margin friction of the 2-nanometer transition nor the structural free cash flow drag of multi-year capital expenditure programs exceeding $60 billion annually. While operational delivery across advanced nodes remains flawless, rigorous fundamental discounted cash flow modeling reveals that capital intensity and working capital expansion cap shareholder value accretion well below prevailing trading levels. Our probability-weighted framework establishes a blended target of $284.15, implying -34.3% upside versus spot and dictating a Hold, Valuation Concerns rating.

HOLD	$284.15	-34.3%	$432.75
Hold, Valuation Concerns	BLENDED TARGET	BLENDED UPSIDE	REFERENCE PRICE
INVESTMENT RATING			

The market remains mesmerized by quarter-over-quarter high-performance computing revenue records and relentless top-line beats, ignoring that aggressive multi-year foundry capital deployment inherently compresses free cash flow conversion. While consensus treats accelerated node migrations as pure margin

accretion, the operational reality demonstrates that initial volume ramps dilute gross margins while elevating working capital absorption across extended supply chains.

Executive summary

Taiwan Semiconductor Manufacturing Co operates as the world's preeminent pure-play semiconductor foundry, serving as the sole critical manufacturing foundation for the global fabless design complex, including tier-one customers such as NVIDIA, AMD, Apple, and Broadcom. The commercial platform has structurally pivoted toward high-performance computing, which now commands 68% of corporate platform mix, leaving consumer smartphone platforms at 21%. Technology leadership remains concentrated in advanced nodes, where 7-nanometer and smaller geometries account for 79% of wafer revenue, while the franchise functions as the monopoly supplier of sub-5nm logic integrated with CoWoS advanced packaging architectures.

Recent operational performance confirms that top-line semiconductor demand across accelerated computing remains extraordinary. In Q3 FY2026, corporate revenue printed at a record $46.72 billion, expanding 41.2% year over year and exceeding the upper bound of management's prior guidance range of $45.8 billion. Operating leverage drove net profit to NT$763.06 billion, up 68.7% year over year, while the Q4 revenue midpoint guide of $53.3 billion obliterated consensus expectations. This acceleration followed Q2 FY2026 revenue of $40.20 billion and Q1 FY2026 revenue of $35.90 billion, demonstrating sequential execution that prompted management to elevate the 2024 to 2029 AI accelerator revenue compound annual growth rate outlook to between 54% and 56%.

The fundamental headwind centers on capital expenditure dynamics and their inevitable drag on cash generation. Capital spending totaled $40.64 billion through the first nine months of FY2026, anchored by Q3 FY2026 outlays of $13.84 billion following the formal raise of full-year capital expenditure expectations to between $60 billion and $64 billion during the second quarter. Furthermore, the early ramp of the 2-nanometer node caused gross margin to compress sequentially to 66.4% in Q3 FY2026 from 67.7% in Q2 FY2026. Working capital lengthening, demonstrated by inventory days rising to 87 from 76 a year earlier and receivable days expanding to 29 from 23, reinforces our quantitative thesis that cash flow conversion will lag operating income expansion.

From a balance sheet perspective, the enterprise retains substantial financial insulation. FY2025 reported cash closed at $2,767.9B against total debt of $1,064.6B, yielding a net cash surplus of $1,703.3B. The business generated $992.4B in free cash flow on revenue of $3,809.1B and EBITDA of $2,742.1B during FY2025, yielding an annual free cash flow margin of 26.1%. Nevertheless, at a market capitalisation of $2,244.5B on 5.19 billion diluted shares, the equity trades at a level that fails to accommodate node depreciation friction, rising geopolitical compliance burdens, and structural overseas fabrication cost premiums.

Operational snapshot

$432.75	$3,809.1B	$992.4B	26.1%	1.37	$432.75
REF. PRICE	FY2025 REVENUE	FY2025 FREE CASH FLOW	FCF MARGIN	EQUITY BETA	REF. CLOSE

Fiscal-year figures as reported in company filings and collected by Cascadia Analytics; ref. price as of 2026-10-29.

The moat is Absolute and Monopolistic

The economic moat protecting Taiwan Semiconductor Manufacturing Co rests upon insurmountable process technology barriers, extreme customer switching costs, and capital scale that cannot be replicated by private or state-backed competitors. As the monopoly supplier of sub-5nm logic with CoWoS advanced packaging, the company manufactures the foundational physical substrate required for high-performance AI accelerators and mobile processors. Flagship silicon architectures designed by NVIDIA, AMD, Apple, and Broadcom are physically co-designed around the company's proprietary design rules, standard cell libraries, and packaging flows, creating prohibitive multi-year switching friction that insulates the order book from competitive displacement.

Pricing power across leading-edge nodes remains demonstrated throughout the financial record. In Q4 FY2025, gross margin reached 62.3% against guidance of 59.0% to 61.0%, which management confirmed was driven by firm wafer pricing and elevated fab utilization. Pricing resilience expanded further into FY2026, where Q1 gross margin printed at 66.2% versus 63.0% to 65.0% guidance, and Q2 gross margin reached 67.7% against 65.5% to 67.5% guided. Even as gross margin compressed to 66.4% in Q3 FY2026 due to the initial depreciation absorption of the 2-nanometer transition, underlying net profit expanded 68.7% year over year to NT$763.06 billion, confirming the franchise's unique capacity to extract economic rents across its customer base.

Customer retention is reinforced by the integration of front-end lithography with backend CoWoS advanced packaging. Fabless designers cannot achieve target thermal efficiency and memory bandwidth solely through silicon scaling, making advanced multi-die packaging mandatory. With 7-nanometer and below representing 79% of total wafer revenue and high-performance computing accounting for 68% of platform mix in Q3 FY2026, customers are tightly tethered to the company's fabrication ecosystem. The velocity of the 2-nanometer ramp, which surged from 3% of wafer revenue in Q2 FY2026 to 8% in Q3 FY2026, highlights that hyperscale customers migrate to TSMC's next-generation architectures at unprecedented speeds, preempting alternative sourcing strategies.

External barriers reinforce this monopolistic position. While geopolitical headwinds emerged in January 2026 via the imposition of 25% Section 232 semiconductor tariffs and ongoing uncertainties surrounding export control postures for advanced computing ICs to China, the

global dependency on Taiwan Semiconductor Manufacturing Co's advanced fabs limits the commercial feasibility of state-sponsored alternatives. Overseas fab buildouts carry structural cost premiums relative to Taiwan operations, yet tier-one clients continue absorbing these operational costs to secure access to sub-5nm capacity, preserving the firm's foundational industry dominance.

Valuation

Three scenarios make the valuation range explicit

Our quantitative valuation methodology implements a multi-scenario discounted cash flow architecture covering the FY2026 through FY2030 forecast window. The framework isolates revenue compound annual growth rates, terminal free cash flow margins, capital intensity normalization, and a dynamic cost of capital across 5.19 billion diluted shares. Baseline revenue compounding across the five-year forecast reflects structural high-performance computing secular trends offset by cyclical pauses, while terminal values are determined via Gordon growth capitalizing normalized free cash flow at an enterprise base terminal growth rate of 3.0%.

The intrinsic valuation model derives a probability-weighted blended target price of $284.15 per share, representing a -34.3% downside variance relative to the reference spot price of $432.75. This output synthesizes three rigorous, probability-weighted scenarios: a 25% bear case allocation yielding $199.67 per share, a 50% base case allocation yielding $282.08 per share, and a 25% bull case allocation yielding $372.79 per share. Across all modeled scenarios, intrinsic enterprise value fails to support prevailing market prices, reflecting a persistent valuation disconnect driven by excessive multiple expansion.

Free cash flow margin trajectory serves as the central operational bridge within our financial engineering. From the FY2025 historical base free cash flow margin of 26.1%, future cash conversion compresses across all forward modeling paths due to elevated capital expenditure programs, overseas operational footprint dilution, and working capital drag. The base case models a 11.6% revenue compound annual growth rate through FY2030, with terminal free cash flow margins settling at 23.0%, while the bear and bull scenarios establish terminal cash conversion margins of 20.9% and 23.5%, respectively.

SCENARIO	WEIGHT	REVENUE CAGR	TARGET FCF MARGIN	TARGET	UPSIDE
Bear	25%	9.0%	20.9%	$199.67	-53.9%
Base	50%	11.6%	23.0%	$282.08	-34.8%
Bull	25%	15.5%	23.5%	$372.79	-13.9%

Bear $199.67

Base $282.08

Bull $372.79

The bear scenario models a 9.0% revenue CAGR through FY2030 and a terminal FCF margin compression to 20.9%, triggered by the combined drag of 25% Section 232 tariffs, tightening export control restrictions to China, and severe overseas fab cost premiums. Free cash flow generation is further constrained as capital expenditures remain elevated near the top of the historical guidance bands while working capital lengthening absorbs operating liquidity, yielding a target of $199.67.

The base scenario models an 11.6% revenue CAGR through FY2030 and a terminal FCF margin of 23.0%, reflecting management's raised long-term AI outlook counterbalanced by unavoidable 2-nanometer depreciation friction. Capital spending stabilizes following the $60 billion to $64 billion FY2026 cycle, allowing normalized operating cash generation to compound over a 5.19 billion share base and producing an intrinsic value of $282.08.

The bull scenario models a 15.5% revenue CAGR and a 23.5% terminal FCF margin, capitalizing on the rapid migration to the 2-nanometer node and realization of the 54% to 56% AI accelerator CAGR. Strong pricing leverage across sub-5nm logic and CoWoS packaging absorbs early node depreciation and overseas cost premiums, generating an intrinsic value of $372.79, which still represents a -13.9% discount to the prevailing spot price.

Valuation framework

A live discount rate, not a static hurdle

Our quantitative valuation architecture rejects static discount thresholds in favor of a market-calibrated Capital Asset Pricing Model derivation, establishing an enterprise weighted average cost of capital of 9.79%. The discount rate incorporates an equity beta input of 1.37, capturing Taiwan Semiconductor Manufacturing Co's structural exposure to global macroeconomic semiconductor cycles, high-performance computing capital expenditure swings among hyperscalers, and elevated geopolitical beta stemming from concentration in the Taiwan Strait. This 9.79% hurdle rate reflects the real rate of return demanded by global institutional allocators to underwrite the combination of technological execution risks and cross-border regulatory friction.

Discounting multi-year cash flows at a 9.79% WACC establishes a rigorous valuation discipline that penalizes back-loaded cash generation. Because the company is deploying unprecedented capital outlays, including $40.64 billion through the first nine months of FY2026 and guided full-year capex of $60 billion to $64 billion, net free cash generation is substantially compressed in the near term. Applying a 9.79% discount factor across the FY2026 to FY2030 forecast window demonstrates that public equity markets at $432.75 are under-discounting terminal duration risk while assuming an unrealistically low cost of capital for a capital-intensive manufacturing asset operating under regulatory scrutiny.

FCF margin normalization is the engine

Free cash flow margin trajectory dictates the divergence between accounting net income and true shareholder cash returns. In FY2025, the company posted a reported free cash flow margin of 26.1%, generating $992.4B in free cash flow from $3,809.1B in revenue. However, operating dynamics recorded throughout recent quarters confirm that peak cash conversion faces sustained downward pressure. In Q3 FY2026, gross margin declined to 66.4% from 67.7% in Q2 FY2026 due to the early-stage depreciation of the 2-

nanometer ramp, while management guided Q4 FY2026 gross margin to a range of 65% to 67%, remaining below previous peaks.

Simultaneously, balance sheet working capital expansion consumes operating liquidity. In Q2 FY2026, inventory days expanded to 87 from 76 a year earlier, while receivable days lengthened to 29 from 23. Combined with structural cost premiums associated with overseas manufacturing facilities, these working capital realities enforce terminal free cash flow margin normalization across all three scenarios. We model terminal free cash flow margins of 20.9% in the bear case, 23.0% in the base case, and 23.5% in the bull case, confirming that even robust top-line execution cannot prevent cash conversion from settling below historical peak levels.

Mechanics are explicit

Revenue becomes EBIT, cash taxes come out, then we add back D&A, subtract CapEx and the change in net working capital to reach unlevered free cash flow, which we discount and divide by diluted shares.

> **Methodology.** Detailed EBIT-bridge DCF on three scenarios, probability-weighted 25/50/25. Beta decays toward a mature peer over the forecast. Terminal value via Gordon growth. Targets are model-computed, not estimated.

KEY VALUATION PARAMETERS

Forecast window	FY2026 to FY2030
Live WACC	9.79% (2026-10-29)
Terminal growth	3.0%
Equity beta	1.37
Diluted shares	5.19 billion

HOLD

over $284.15 blended target

Competitive moat and concentration risk

We rate the moat as Wide and we view the market structure as Near-monopoly at leading edge; oligopoly overall foundry. We assign a premium of +30% to sector median multiples on moat strength alone.

- foundry scale plus process lead, 3nm at sustained full utilization, 2nm ramping to 8% of revenue in a single quarter (Q3 2026, TechTimes October 15, 2026), and CoWoS advanced packaging with spillover demand.
- IDC (2026), TSMC raised wafer pricing by more than 5% while raising its 3nm monthly capacity target to 165,000 wafers and CoWoS to 125,000 wafers, on sustained full utilization, raising price into full utilization is the cleanest pricing-power test in this batch.

Corroborating, 2nm wafers at ~$30,000, roughly 50% above 3nm pricing, with demand unabated (TechTimes, October 15, 2026).

- The weakest element is concentration, high-performance computing (AI accelerators) was 68% of Q3 2026 revenue (+22% sequentially), and essentially all leading-edge capacity sits in Taiwan, so geopolitical and single-cycle concentration cut against the top of the band.

On a multiple cross-check, the justified range is 30.0-33.8x / 22.9-25.8x. The cross-check value is $224.0-$252.3 (pt $242.6), on a base of $9.78/ADR (49.328% normalized EBIT × FY2025 revenue $122.42bn × 0.84 ÷ 5,186.5M ADRs). The gap verdict is SHRINKS, in part small part ~11% at point (2-16% range).

This figure uses the normalized base from the October 27 recheck, not the original boom-year base.

We have no disclosed backlog figure for this company, so backlog gives no support to the growth path here.

We score concentration and credit risk across seven families, where a High score is 8 points, a Medium score is 4 points, and a Low score is 1 point.

Risk family	Score
customer concentration	8 points
counterparty credit and funding dependence	1 points
single product, single program, or single platform dependency	8 points
single supplier, single fab, or single geography exposure	8 points
platform, channel, or distribution dependency	0 points
customer insourcing risk	4 points
regulatory single point risk	8 points

The combined discount is 20 percent, capped at 20 percent, against an uncapped total of 37 percent. Shown side by side with the moat premium or discount, the net stance is +10 pp points arithmetically. We do not net these figures into a new target, and we do not double count a family excluded under the double-count rule.

The ladder ranks risk coarsely, and because 32 of the 47 names reach the 20 percent cap, the combined discount is not a mechanical valuation haircut until the ladder is recalibrated.

On the gap to spot, the concentration assessment reads as follows. Not at all (corrected from batch arithmetic)

Both the moat premium or discount and the concentration discount are multiple and risk cross-checks dated October 27, 2026, and neither figure is in the DCF target itself. Cross-check figures were computed against the Base in the October 27 framework, so they may differ from a live Base that has since moved.

Business & segment drivers

Revenue prospects anchor on High-Performance Computing Acceleration and Advanced Node Migration HOLD

Corporate expansion is overwhelmingly governed by the explosive growth of the high-performance computing platform, which expanded 22% sequentially in Q3 FY2026 to reach 68% of total revenue. This structural shift represents a permanent realignment from historical consumer electronics cycles, where smartphone platforms accounted for dominant volume. In Q3 FY2026, smartphones contracted to 21% of revenue mix, down from 22% in Q2 FY2026 and 32% in Q4 FY2025. The high-performance computing mix is driven entirely by hyperscale AI accelerator demand from fabless designers including NVIDIA, AMD, and Broadcom, validating management's raised multi-year outlook targeting a 54% to 56% AI accelerator revenue compound annual growth rate through 2029.

The secondary revenue engine is the accelerated commercialization of sub-5nm logic architectures. Wafer revenue from advanced nodes at 7-nanometer and below reached 79% in Q3 FY2026, expanding from 77% in Q2 FY2026 and 74% in Q1 FY2026. Within this mix, the 2-nanometer node demonstrated unprecedented adoption velocity, leaping from an initial 3% of wafer revenue in Q2 FY2026 to 8% in Q3 FY2026. This rapid ramp outpaces any previous node transition in foundry history, driven by fabless customers demanding immediate density, performance, and power efficiency gains to support next-generation compute clustering.

Advanced packaging serves as the indispensable companion to front-end node transitions. Taiwan Semiconductor Manufacturing Co's proprietary CoWoS packaging technology forms a technological monopoly for packaging high-bandwidth memory adjacent to leading-edge compute dies. Because advanced logic cannot function in modern AI data centers without high-bandwidth interconnects, CoWoS capacity allocations dictate customer shipment volumes. This integration provides the enterprise with unmatched commercial leverage, enabling the company to maintain high fab utilization and secure pricing power across its entire sub-5nm logic portfolio.

However, this structural growth profile demands extraordinary capital re-investment. FY2026 capital expenditure was guided upward to $52 billion to $56 billion in Q1 FY2026, up 27% to 40% from $40.9 billion in 2025, and was raised further in Q2 FY2026 to a range of $60 billion to $64 billion. With $40.64 billion deployed through the first nine months of FY2026, capital intensity suppresses immediate free cash flow generation. Coupled with inventory days stretching to 87 days and trade receivables expanding to 29 days, the business is reinvesting operating cash flows back into fab tooling and working capital at an unprecedented scale.

What to monitor

- **2-Nanometer Gross Margin Trajectory** Track whether gross margin can rebound from 66.4% in Q3 FY2026 and guided 65% to 67% in Q4 FY2026 toward the 67.7% peak printed in Q2 FY2026 as N2 volume yields mature.
- **Capital Expenditure Discipline vs Guidance Bands** Monitor full-year FY2026 outlays against the raised $60 billion to $64 billion forecast following $40.64 billion deployed through nine months, watching for capital intensity easing into FY2027.
- **Working Capital and Supply Chain Absorption** Evaluate whether inventory days normalize from 87 days and receivable days contract from 29 days as recorded in Q2 FY2026.
- **Geopolitical and Trade Compliance Pressures** Assess operational adaptations surrounding the 25% Section 232 semiconductor tariffs enacted in January 2026 alongside evolving US export controls governing advanced computing IC shipments to China.

+31.6%	59.9%	50.8%	26.1%	$1,703.3B	31.5%
FY2025 REVENUE GROWTH	GROSS MARGIN	OPERATING MARGIN	FCF MARGIN	NET CASH	RETURN ON EQUITY

METRIC	FY2025 (AS REPORTED)
Revenue	$3,809.1B
Gross Profit	$2,281.3B
EBIT	$1,936.1B
Net Income	$1,697.6B
EBITDA	$2,742.1B
Free Cash Flow	$992.4B
Diluted EPS	$331.25
ROE	31.5%

Risk & moat matrix

The bear case

SHARE EROSION	M&A DIGESTION	REGULATORY AND PRICING SCRUTINY	TECHNOLOGY DISRUPTION	REINVESTMENT TREADMILL	WACC AND DURATION SENSITIVITY
While TSMC holds a	TSMC expands via		Ramping next-generation	Capital intensity has escalated	

monopoly supplier position in sub-5nm logic, any technological yield maturity from alternate foundries could dilute pricing power across legacy advanced nodes. Given that 7-nanometer and below constitutes 79% of wafer revenue, even modest customer volume dual-sourcing would pressure fab utilization.

pure-play organic fab construction rather than large-scale corporate mergers, avoiding traditional balance sheet goodwill write-downs. However, integrating complex international fab infrastructure operations represents an equivalent operational execution risk across diverse labor and supply environments.

The imposition of 25% Section 232 semiconductor tariffs in January 2026 highlights intensifying cross-border trade friction that could disrupt global silicon distribution networks. Furthermore, shifting export control postures regarding advanced computing integrated circuits to China threatens total addressable demand across hyperscale platforms.

geometries introduces extreme engineering vulnerabilities where yield degradation directly impacts corporate profitability. As observed in Q3 FY2026, early 2-nanometer depreciation diluted gross margin to 66.4% from 67.7%, confirming that node complexity exerts severe margin drag prior to yield optimization.

dramatically, with full-year FY2026 capital expenditure raised to between $60 billion and $64 billion following $40.64 billion deployed through nine months. This continuous reinvestment treadmill severely suppresses free cash flow conversion despite record operating profits, leaving minimal cash for non-operating distributions.

With an equity beta of 1.37 and a quantitative WACC of 9.79%, the enterprise valuation exhibits acute sensitivity to long-term discount rates. Back-loaded terminal cash flows lose substantial present value under this hurdle rate, rendering the prevailing spot price of $432.75 highly vulnerable to multiple contraction.

The strengths

OLIGOPOLY POSITION

Taiwan Semiconductor Manufacturing Co commands an effective monopoly as the sole pure-play foundry capable of high-volume sub-5nm logic fabrication

FCF COMPOUNDING

The enterprise generated $992.4B in free cash flow during FY2025 on a robust 26.1% FCF margin, supported by an operating cash balance of $2,767.9B. This immense self-

GEOGRAPHIC DURABILITY

The core concentration of fabrication clusters across Taiwan delivers unmatched operational, logistical, and yield efficiencies that overseas facilities cannot replicate.

PIPELINE OPTIONALITY

The commercial pipeline is underpinned by management's upgraded 2024 to 2029 AI accelerator revenue CAGR outlook of 54% to 56%. The explosive adoption of the 2-

integrated with CoWoS advanced packaging. Fabless leaders such as NVIDIA, AMD, Apple, and Broadcom remain entirely dependent upon this single manufacturing architecture.

funding operational scale ensures that TSMC can underwrite record capital expenditure programs without compromising its pristine net cash position of $1,703.3B.

This tightly integrated domestic ecosystem sustains superior fab utilization rates and enabled record corporate operating margins of 60.3% in Q2 FY2026.

nanometer node, surging to 8% of wafer mix in Q3 FY2026, guarantees multi-year revenue visibility across leading-edge high-performance computing.

The bottom line on risk and reward

Taiwan Semiconductor Manufacturing Co stands as an operational titan, commanding the commanding heights of the global technology economy. Its commercial execution across the first nine months of FY2026 has been exceptional, highlighted by Q3 revenue printing at a record $46.72 billion, net profits expanding 68.7% year over year to NT$763.06 billion, and management issuing an extraordinary Q4 revenue midpoint guidance of $53.3 billion. The franchise's technological lead in sub-5nm logic, CoWoS advanced packaging, and the accelerated 2-nanometer adoption curve guarantee that fabless design leaders will remain tied to its manufacturing footprint for the remainder of the decade.

Fundamental equity valuation, however, cannot divorce operational excellence from price paid per share. At the reference spot price of $432.75, public equity markets are capitalizing peak operating performance without accounting for the capital requirements mandated to sustain it. Raising FY2026 capex to between $60 billion and $64 billion, absorbing 2-nanometer initial depreciation that contracted gross margins to 66.4% in Q3, managing extended working capital cycles of 87 inventory days, and navigating 25% Section 232 tariffs impose structural caps on free cash flow generation. Our probability-weighted discounted cash flow model explicitly prices these cash flow realities across 5.19 billion diluted shares at a 9.79% WACC.

The quantitative framework establishes a blended target price of $284.15 per share, derived from our 25% bear ($199.67), 50% base ($282.08), and 25% bull ($372.79) scenarios. This blended target reveals a -34.3% downside variance relative to prevailing market prices. While long-term business fundamentals remain unmatched across global technology, equity pricing reflects unbridled perfection that leaves zero margin of safety against execution friction or valuation mean-reversion. We reiterate our Hold, Valuation Concerns rating, advising institutional allocators to await a substantial valuation reset before initiating or expanding equity exposure.

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Franchise deep dive

Sub-5nm Logic and CoWoS Advanced Packaging: The Monopoly Platform

The structural core of Taiwan Semiconductor Manufacturing Co's economic franchise resides in its undisputed monopoly over sub-5nm logic fabrication integrated with CoWoS advanced packaging. Advanced computing silicon has migrated beyond the boundaries of standard planar lithography, requiring heterogeneous chiplet architectures that combine high-density compute dies with high-bandwidth memory stacks. Because TSMC commands the only commercially viable, high-volume production line capable of fabricating sub-5nm logic integrated via CoWoS architectures, global fabless computing leaders including NVIDIA, AMD, Apple, and Broadcom possess no alternative manufacturing source capable of meeting their volume and yield specifications.

This structural monopoly is clearly visible in the platform mix. In Q3 FY2026, high-performance computing accounted for 68% of total corporate revenue, expanding 22% sequentially and completely eclipsing traditional mobile platforms, which fell to 21% of revenue. Advanced nodes at 7-nanometer and below generated 79% of total wafer revenues during the third quarter, confirming that leading-edge architectures capture an overwhelming share of capital allocation. Advanced node revenue dominance expanded consistently from 74% in Q1 FY2026 (where 3-nanometer alone represented 25%) and 77% in Q2 FY2026, demonstrating that hyperscale data center infrastructure spending flows directly through TSMC's manufacturing lines.

The commercialization velocity of the 2-nanometer node reinforces the strength of this operational chokehold. Introduced into commercial production during Q2 FY2026 at 3% of wafer revenue, the 2-nanometer node ramped to 8% of wafer revenue in Q3 FY2026, marking the steepest node adoption ramp in corporate history. Tier-one customers are transitioning their next-generation architectures to N2 significantly faster than observed in prior technology generations, motivated by severe compute density requirements. This rapid transition cements pricing authority, enabling the company to guide Q4 FY2026 revenue to an unprecedented midpoint of $53.3 billion, representing roughly 14% sequential top-line expansion.

Nonetheless, the physical reality of maintaining this monopoly imposes massive financial burdens. Generating sub-5nm wafers requires continuous capital outlays, as evidenced by FY2026 capex rising from an initial guidance band of $52 billion to $56 billion up to $60 billion to $64 billion, with $40.64 billion already expended through nine months. Furthermore, the rapid onboarding of 2-nanometer capacity introduced early-stage depreciation friction, pulling gross margins down from 67.7% in Q2 FY2026 to 66.4% in Q3 FY2026, with Q4 guided to 65% to 67%. While the franchise moat remains absolute, the capital required to sustain it restricts cash flow conversion, reinforcing our valuation thesis.

What the recent record shows

- In Q3 FY2026, high-performance computing expanded 22% sequentially to reach 68% of platform revenue, while the 2-nanometer node advanced from 3% to 8% of total wafer revenue in a single quarter.
- Q2 FY2026 delivered record financial performance with gross margin printing at 67.7% and operating margin reaching 60.3%, driving net income up 77.4% year over year to NT$706.56 billion.
- Management significantly elevated capital deployment plans during FY2026, raising full-year capital expenditure guidance to $60 billion to $64 billion while expanding the 2024 to 2029 AI accelerator revenue CAGR outlook to 54% to 56%.
- Working capital cycles lengthened during Q2 FY2026, with inventory days expanding to 87 from 76 a year earlier and receivable days rising to 29 from 23, signaling cash absorption across the extended global supply chain.

Our read: Taiwan Semiconductor Manufacturing Co's monopoly over sub-5nm logic and CoWoS advanced packaging guarantees its role as the indispensable foundation of global computing. However, extreme capital intensity, 2-nanometer depreciation dilution, and lengthening working capital cycles restrain free cash flow conversion, creating an insurmountable valuation disconnect at prevailing trading prices.